

```
select * from customer limit 20
```

--Q1. What is the total revenue generated by male vs. female customers?

```
select gender, SUM(purchase_amount) as revenue
from customer
group by gender
```

--Q2. Which customers used a discount but still spent more than the average purchase amount?

```
select customer_id, purchase_amount
from customer
where discount_applied = 'Yes' and purchase_amount >= (select AVG(purchase_amount)
from customer)
```

--Q3. Which are the top 5 products with the highest average review rating?

```
select item_purchased, ROUND(AVG(review_rating::numeric),2) as "Average Product
Rating"
from customer
group by item_purchased
order by avg(review_rating) desc
limit 5
```

--Q4. Compare the average Purchase Amounts between Standard and Express Shipping.

```
select shipping_type,
ROUND(AVG(purchase_amount),2)
from customer
where shipping_type in ('Standard','Express')
group by shipping_type
```

--Q5. Do subscribed customers spend more? Compare average spend and total revenue between subscribers and non-subscribers.

```
select subscription_status,
COUNT(customer_id) as total_customers,
ROUND(AVG(purchase_amount),2) as avg_spend,
ROUND(SUM(purchase_amount),2) as total_revenue
from customer
group by subscription_status
order by total_revenue, avg_spend desc
```

--Q6. Which 5 products have the highest percentage of purchases with discounts applied?

-- this query shows that which products are brought when discounts applied

```
select item_purchased,
ROUND(100 * SUM(CASE WHEN discount_applied = 'Yes' THEN 1 ELSE 0 END)/
COUNT(*),2) as discount_rate
from customer
group by item_purchased
```

```
order by discount_rate desc
limit 5
```

--Q7. Segment customers into New, Returning, and Loyal based on their total number of previous purchases, and show the count of each segment.

```
with customer_type as(
select previous_purchases,customer_id,
CASE
    WHEN previous_purchases = '1' THEN 'New'
    WHEN previous_purchases BETWEEN 2 and 10 THEN 'Returning'
    ELSE 'Loyal'
END AS customer_segment
from customer
)
select customer_segment, count(*) as "Number of customers"
from customer_type
group by customer_segment
```

--Q8. What are the top 3 most purchased products within each category?

```
-- CTE / TEMP table
with item_counts as (
select category,item_purchased,
COUNT (customer_id) as total_orders,
--ROW_NUMBER()Give each row a number (1,2,3...)
--PARTITION BY categoryRestart numbering for each category
ROW_NUMBER() over (partition by category order by count(customer_id)DESC) as
item_rank
from customer
group by category,item_purchased
)
```

```
select item_rank,category,item_purchased,total_orders$
from item_counts
where item_rank<=3
```

--Q9. Are customers who are repeat buyers (more than 5 previous purchases) --also likely to subscribe?

```
select subscription_status,
COUNT(customer_id) as repeat_buyers
from customer
WHERE previous_purchases >=5
group by subscription_status
```

--Q10. What is the revenue contribution of each age group?

```
select age_group, SUM(purchase_amount) as total_revenue
```

```
from customer  
group by age_group  
order by total_revenue DESC
```

```
select * from customer
```